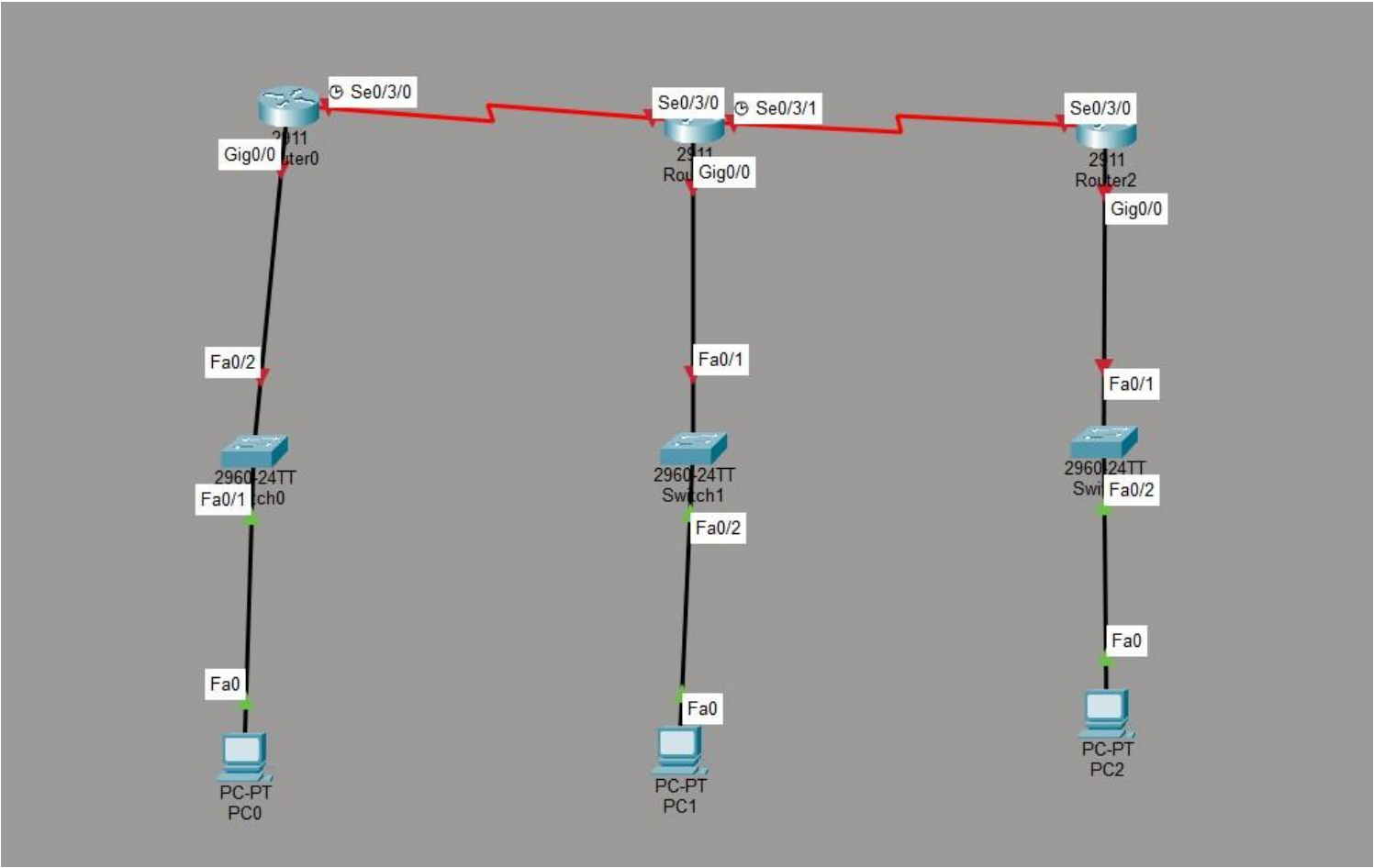


Task-1

Set up a network topology with three routers in Cisco Packet Tracer, connect them using serial or Ethernet links, and assign IP addresses to all interfaces.

To design and configure a network topology containing three Cisco routers in Cisco Packet Tracer, connect the routers using serial links, connect each router to a LAN, assign IPv4 addresses to all required interfaces and end devices, configure routing, and verify connectivity between the networks.

Network Topology



IP Addressing Scheme

Device	Interface	IP Address	Subnet Mask	Purpose
Router0	G0/0	192.168.1.1	255.255.255.0	LAN 1 gateway
Router0	S0/3/0	10.0.0.1	255.255.255.252	Link to Router1
Router1	G0/0	192.168.2.1	255.255.255.0	LAN 2 gateway
Router1	S0/3/0	10.0.0.2	255.255.255.252	Link to Router0
Router1	S0/3/1	10.0.0.5	255.255.255.252	Link to Router2
Router2	G0/0	192.168.3.1	255.255.255.0	LAN 3 gateway
Router2	S0/3/0	10.0.0.6	255.255.255.252	Link to Router1
PC0	Fa0	192.168.1.10	255.255.255.0	End device
PC1	Fa0	192.168.2.10	255.255.255.0	End device
PC2	Fa0	192.168.3.10	255.255.255.0	End device

Default gateways:
PC0 → 192.168.1.1
PC1 → 192.168.2.1
PC2 → 192.168.3.1

Router Configuration Router0

```

Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router(config)#ip route 192.168.3.0 255.255.255.0 10.0.0.2
Router(config)#do wr
Building configuration...
[OK]
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#enable
Router#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.1.1	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Serial0/3/0	unassigned	YES	unset	administratively down	down
Serial0/3/1	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

```

Router#enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interace serial 0/3/0
^
% Invalid input detected at '^' marker.

Router(config)#interface serial 0/3/0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Router(config)#exit

```

Router 1

```
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet 0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface serial 0/3/0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router(config-if)#exit
Router(config)#interface serial 0/3/1
Router(config-if)#ip address 10.0.0.5 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down
Router(config-if)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#ip route 192.168.3.0 255.255.255.0 10.0.0.6
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router 2

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet 0/0
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface serial 0/3/0
      ^
% Invalid input detected at '^' marker.

Router(config)#interface serial 0/3/0
Router(config-if)#ip address 10.0.0.6 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
do wr
Building configuration...
[OK]
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.5
Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.5
Router(config)#do wr
Building configuration...
[OK]
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Verification and Testing

```
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126

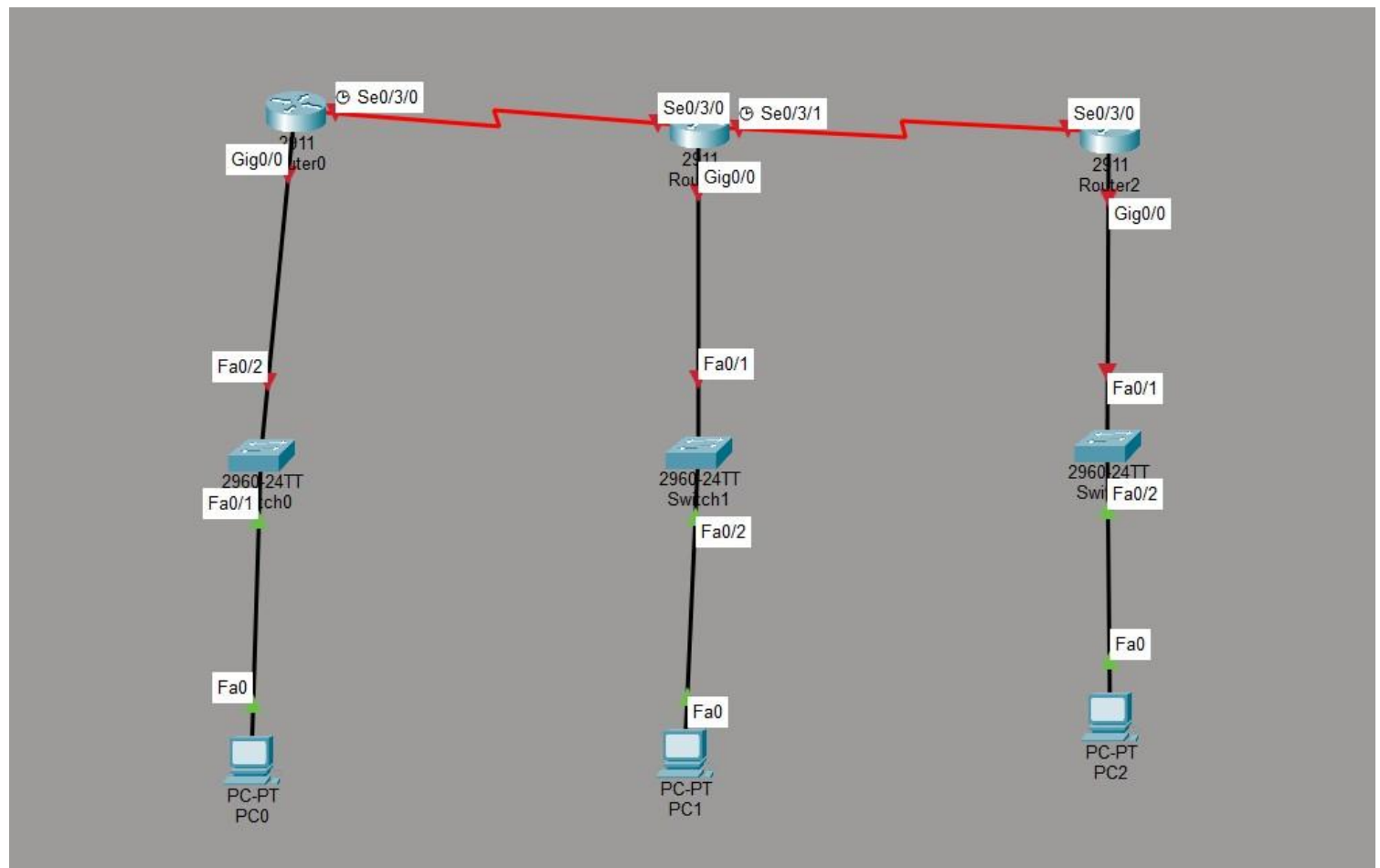
Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>
```

Task -2

Enable RIP version 2 on all three routers in your topology, advertise all directly connected networks, and verify that each router learns routes to all subnets using the 'show ip route' command.

To configure a three-router network in Cisco Packet Tracer, assign IPv4 addresses to router and PC interfaces, connect routers using serial links, enable RIP version 2 on all routers, advertise all directly connected networks, and verify learned routes using show ip route. **Network Topology**



IP Addressing Table

Device	Interface	IP Address	Subnet Mask	Purpose
Router0	G0/0	192.168.1.1	255.255.255.0	LAN 1 gateway
Router0	S0/3/0	10.0.0.1	255.255.255.252	Router0-Router1
Router1	G0/0	192.168.2.1	255.255.255.0	LAN 2 gateway
Router1	S0/3/0	10.0.0.2	255.255.255.252	Router1-Router0
Router1	S0/3/1	10.0.0.5	255.255.255.252	Router1-Router2
Router2	G0/0	192.168.3.1	255.255.255.0	LAN 3 gateway
Router2	S0/3/0	10.0.0.6	255.255.255.252	Router2-Router1
PC0	Fa0	192.168.1.10	255.255.255.0	Gateway 192.168.1.1

PC1	Fa0	192.168.2.10	255.255.255.0	Gateway 192.168.2.1
PC2	Fa0	192.168.3.10	255.255.255.0	Gateway 192.168.3.1

RIP Version 2 Configuration

Router 0

```
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.1.0
Router(config-router)#network 10.0.0.0
Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

Router 1

```
Router(config)#
Router(config)#router rip
Router(config-router)#
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.2.0
Router(config-router)#network 10.0.0.0
Router(config-router)#network 10.0.0.4
Router(config-router)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router 2

```
Router(config)#
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.3.0
Router(config-router)#network 10.0.0.4
Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Verification

Use these commands after RIP has converged:

show ip interface brief show ip route show ip protocols

In show ip route, C indicates a directly connected network and R indicates a route learned through RIP.

Router	RIP-learned networks expected	Verification
Router0	192.168.2.0/24, 10.0.0.4/30, 192.168.3.0/24	R entries
Router1	192.168.1.0/24, 192.168.3.0/24	R entries
Router2	10.0.0.0/30, 192.168.1.0/24, 192.168.2.0/24	R entries

Connectivity Testing

PC 0

```
C:\>ping 192.168.3.10

Pinging 192.168.3.10 with 32 bytes of data:

Reply from 192.168.3.10: bytes=32 time=9ms TTL=125
Reply from 192.168.3.10: bytes=32 time=2ms TTL=125
Reply from 192.168.3.10: bytes=32 time=3ms TTL=125
Reply from 192.168.3.10: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.3.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 9ms, Average = 4ms

C:\>
```

RIP version 2 was enabled on all three routers and all directly connected LAN and serial networks were advertised. After convergence, remote networks are learned dynamically and appear with R in the routing table.

Task-3

RIP Hop Count Change Experiment

Objective

To observe how adding an extra router changes the **RIP hop-count metric** for a destination network.

Steps

1. Check the original route

On the router, run:

show ip route

Find the RIP route. For example:

R 192.168.3.0/24 [120/2] via 192.168.2.2

Here, **2** is the RIP hop count.

2. Add an extra router

Suppose the original topology is:

```
R1 ----- R2 ----- R3
                |
                192.168.3.0/24
```

Add **R4** between R2 and R3:

```
R1 ----- R2 ----- R4 ----- R3
                |
                192.168.3.0/24
```

Configure the new links and enable RIP on the added router so that the network remains reachable.

3. Check the routing table again

On R1:
show ip route
The route may now appear similar to:
R 192.168.3.0/24 [120/3] via 192.168.2.2

Observation

Condition	RIP Route	Hop Count
Before adding R4	192.168.3.0/24	2 hops
After adding R4	192.168.3.0/24	3 hops

Result

Adding one router to the path increases the RIP metric by **one hop**. Therefore, a route that previously had a metric of **2** can become **3**, provided the new router is actually part of the selected RIP path.

Conclusion: RIP uses **hop count** as its routing metric. Each additional router that a packet must pass through increases the RIP hop count by one.

Task-4

Use the simulation mode in Cisco Packet Tracer to capture and analyze RIP update packets as they are exchanged between routers.

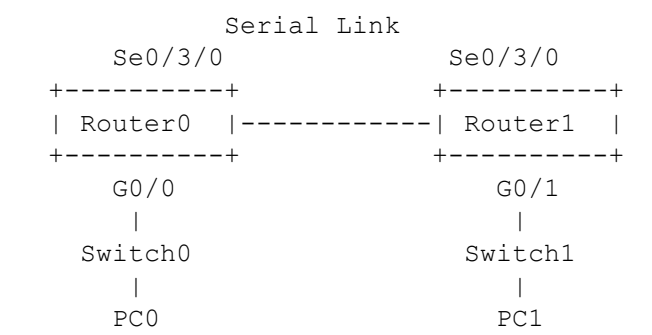
Hint: Filter for RIP packets in the simulation panel and note the source, destination, and advertised routes in each update.

To use the **Simulation Mode** in Cisco Packet Tracer to capture and analyze **RIP (Routing Information Protocol) update packets** exchanged between two routers and identify the source, destination, advertised routes, and routing metrics.

- To configure a network with two routers.
- To understand the operation of RIP.
- To observe RIP update packets in Cisco Packet Tracer Simulation Mode.
- To identify the source and destination of RIP packets.
- To examine the routes advertised by each router.

To understand how routers exchange routing information dynamically.

Topology



Configuration

RIP was configured on both routers so that they could exchange routing information.

Router0

```
Router0(config)# router rip
Router0(config-router)# version 2
Router0(config-router)# network <Router0-LAN-Network>
Router0(config-router)# network <Serial-Network>
Router0(config-router)# no auto-summary
```

Router1

```
Router1(config)# router rip
Router1(config-router)# version 2
Router1(config-router)# network <Router1-LAN-Network>
Router1(config-router)# network <Serial-Network> Router1(config-router)# no auto-summary
```

The actual network addresses depend on the IP addressing used in the Packet Tracer topology.

7. Procedure

- 1. Created the network topology using two routers, two switches, and two PCs.
- 2. Connected Router0 and Router1 using their serial interfaces.
- 3. Connected each router to a switch.
- 4. Connected each switch to a PC.
- 5. Configured IP addresses on the router interfaces and PCs.
- 6. Configured RIP version 2 on both routers.
- 7. Verified that the router interfaces were operational.
- 8. Opened **Simulation Mode** in Cisco Packet Tracer.
- 9. Selected **Edit Filters**.
- 10. Removed unnecessary protocols and selected **RIP**.
- 11. Used **Capture/Forward** to observe the packets exchanged between Router0 and Router1.
- 12. Opened the RIP packet and examined the PDU details.
- 13. Recorded the source, destination, advertised networks, and metric.

8. RIP Packet Analysis

During the simulation, RIP update packets were observed traveling between the two routers. **RIP Update from Router0 to Router1**

Parameter	Observation
Protocol	RIP
Source	Router0
Destination	Router1 / RIP destination
Advertised Route	<Router0-LAN-Network>
Metric	<Metric shown in PDU>

Router0 advertised the network connected to its LAN so that Router1 could learn how to reach that network.

RIP Update from Router1 to Router0

Parameter	Observation
Protocol	RIP
Source	Router1
Destination	Router0 / RIP destination
Advertised Route	<Router1-LAN-Network>
Metric	<Metric shown in PDU>

Router1 advertised its LAN network so that Router0 could learn how to reach the network connected to Router1.

Simulation Observation

The RIP packets were successfully observed in **Simulation Mode**.

The general packet exchange was:

Router0 ——— RIP Update ———▶ Router1

Router1 ——— RIP Update ———▶ Router0

The routers exchanged information about the networks that they knew about. The advertised routes were used by the receiving router to update its routing table.

RIP is a **distance-vector routing protocol**.

RIP uses **hop count** as its routing metric.

- A directly connected network has a metric of **0** from the router's perspective.
- A route learned through one neighboring router normally has a hop count of **1**.
- The maximum usable RIP hop count is **15**.
- A hop count of **16 represents an unreachable network**.

RIP version 2 also supports classless routing information and uses multicast address **224.0.0.9** for RIP updates.

RIP update packets were successfully captured and analyzed in Cisco Packet Tracer Simulation Mode. The packets were exchanged between Router0 and Router1 and contained information about reachable networks. Router0 advertised its connected LAN network to Router1, while Router1 advertised its connected LAN network to Router0. RIP uses hop count as its routing metric, and the receiving routers use the advertised information to update their routing tables. Therefore, the experiment successfully demonstrated dynamic route exchange using RIP.